

Estimating the lubricant viscosity ratio in ball bearings with ultrasound and acoustic emission sensor fusion

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Abstract

This paper investigates whether acoustic emission (AE) and passive ultrasound (US) provide complementary information about the lubrication condition of rolling-element bearings. Wideband AE and heterodyned passive US are acquired simultaneously and time-synchronised from a single ball bearing on an instrumented test rig, while the operating condition is varied over a staircase of rotational speeds repeated across a sequence of temperature blocks, driving the lubricant through the boundary, mixed, and full-film regimes. The ISO 281 viscosity ratio κ , computed from the operating condition, serves as a continuous reference for lubrication adequacy. Frequency sub-band features from both channels are regressed onto κ with a regularised linear model and a gradient-boosted tree ensemble, evaluated under an acquisition-grouped protocol that prevents leakage of near-duplicate segments across the training–evaluation boundary. Either channel alone supports accurate κ regression. Combining them significantly outperforms either alone, for both model families under paired group-level tests, providing, to the authors’ knowledge, the first quantitative demonstration of AE–US complementarity. This complementarity is supported with a correlation analysis and by grouped SHAP attribution. These show that the channels encode partly distinct operating-condition information, with US associated with temperature substantially more strongly than AE, and that features from both channels are important for κ regression.

Contents

1	Introduction	5
2	Background	7
2.1	Lubrication regimes and the viscosity ratio κ	7
2.2	Acoustic emission and passive ultrasound for lubrication monitoring	7
3	Experimental Setup	8
3.1	Experimental rig	8
3.2	Experimental protocol for varying κ	9
3.3	Acquisition segments and measured operating conditions	9
4	Signal Processing and Feature Extraction	11
4.1	Waveforms and spectrograms	11
4.2	Sub-band decomposition	12
4.3	Feature set	13
5	Feature Analysis	14
5.1	Correlation with κ	14
5.2	Marginal and conditional correlation with operating conditions	15
6	Regression Modelling and Statistical Evaluation	16
6.1	Experimental design	16
6.2	Model families	17
6.3	Statistical significance testing	17
7	Results	18
7.1	Overall performance	18
7.2	Sensor comparison	19
7.3	Sensor fusion	19
7.4	Clustered SHAP feature importance	20
8	Discussion	21
8.1	AE-US complementarity and channel encoding	21
8.2	On the use of κ as lubrication reference	22
8.3	Evaluation protocol and comparability with prior work	22
9	Conclusion	23

Todo list

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■ Check that this is correct	8
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■ State the resonance frequency and heterodyne specification from the Ultra-Trak 750 datasheet.	9
■ Check that this is true after new dataset.	9
■ Include figure which illustrates the protocol for varying operating conditions and kappa once the speed issue has been settled.	9
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■ Finalize spectrograms with correct colors, ranges, and dashed lines for sub-bands	12
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Nomenclature

Symbols

κ	ISO 281 viscosity ratio, ν/ν_1
ν	kinematic viscosity at operating temperature [$\text{mm}^2 \text{s}^{-1}$]
ν_1	reference viscosity for full film formation [$\text{mm}^2 \text{s}^{-1}$]
n	rotational speed [rpm]
d_{pw}	mean pitch diameter [mm]
N	samples per segment
x_n	voltage sample n
S_k, f_k	one-sided FFT magnitude and frequency at bin k
f_c	spectral centroid [Hz]
σ_w	magnitude-weighted spectral bandwidth [Hz]
ρ	Spearman correlation coefficient

Abbreviations

AE	acoustic emission
CV	cross-validation
DM	Diebold–Mariano
EHL	elastohydrodynamic lubrication
LCM	lubrication condition monitoring
MAE	mean absolute error
RMSE	root mean square error
SHAP	SHapley Additive exPlanations
US	passive ultrasound
VFD	variable-frequency drive

1 Introduction

Rolling-element bearings are among the most widely used components in rotating machinery, and their reliable and efficient operation depends critically on adequate lubrication. When a bearing is in operation, the lubricant forms a thin separating film between the rolling elements and raceways, reducing friction, heat generation, and wear. When the film-forming conditions are inadequate – because of insufficient lubricant quantity, unsuitable viscosity, contamination, or degradation – metal-to-metal asperity contact increases, accelerating surface damage and structural fault development, and ultimately causing premature bearing failure. Lubrication condition monitoring (LCM) therefore addresses an industrially important problem: if lubrication adequacy could be assessed online during operation, and without interfering with the running of the bearing, relubrication could be performed according to the actual operating condition rather than by fixed schedules that may under- or over-lubricate [?]. Developing a non-invasive method that can provide continuous information about lubrication adequacy under changing operating conditions, however, remains challenging.

The established condition-monitoring methods used in rotating machinery are primarily designed for detecting structural faults rather than assessing lubrication adequacy. Vibration analysis, the dominant approach in rotating machinery diagnostics, is effective for structural faults such as raceway spalling and rolling-element damage, where characteristic-frequency signatures can be related to bearing geometry and rotational speed, and where faults cause significant structural vibrations [?]. Lubrication-related changes are, however, more difficult to isolate, since they are weaker, broadband, and non-stationary, and they may be masked by structural resonances, incipient fault signatures, and operating-condition effects. Recently developed data-driven methods can recover lubrication states from high-frequency vibration, but performance is tied to operating conditions such as rotational speed — the same factors that complicate LCM more broadly [? ? ?]. Temperature monitoring, lubricant sampling, optical methods, and reflective ultrasound methods provide useful complementary information, but they are delayed, invasive, or difficult to apply continuously to a bearing in operation [? ? ?]. These limitations motivate the development of non-invasive, online sensing methods that are sensitive to the contact- and friction-related changes associated with variations in lubrication conditions.

Acoustic emission (AE) and passive¹ ultrasound (US) are two non-invasive sensing methods that have shown particular promise in this setting. Both can be measured by sensors mounted externally on the bearing housing, require no access to the lubricant, and can be retrofitted to machines already in service. They arise from the same contact-related sources and so are physically tied to the state of the lubrication film [? ?].

Previous studies provide evidence that both methods are sensitive to lubrication-related changes in rolling-element bearings. AE energy and pulse-count measures have been shown to track lubrication condition [? ? ?], AE responds earlier than vibration to lubricant contamination and structural faults [? ? ?], and changes in the frequency content and power of AE signals have been related to lubrication condition and film thickness through both data-driven methods [?] and physical models [? ?]. Passive US is already used commercially: handheld relubrication probes compare the received signal level against a baseline and indicate relubrication need once a threshold is exceeded [?]. Beyond such threshold tools, however, passive US has received far less attention than AE, with most quantitative work limited to simple acoustic features from low-cost sensors [?].

Despite these promising results, AE- and US-based approaches share important limitations. Acoustic signals are not determined by lubrication alone but are affected by rotational speed, load,

¹Passive methods use signals inherently emanating from the bearing structure in contrast to active methods, which induce energy into the bearing structure and then measure the reaction [?].

temperature, structural transmission paths, sensor coupling, and other operating conditions [? ? ?]. Available datasets also often contain limited variation in operating conditions, making it difficult to assess whether acoustic models generalise beyond the specific conditions under which they were trained. In addition, no universally accepted physically motivated or data-driven reference for instantaneous lubrication conditions exists, making it difficult to benchmark sensing data against a suitable target.

The ISO 281 viscosity ratio κ is instead a commonly used continuous, physically motivated estimate of lubrication conditions. It is calculated from rotational speed, temperature, lubricant viscosity, and bearing geometry and is easily interpretable. However, it does not account for lubricant starvation, degradation, or contamination. Although κ has shortcomings, it can serve as a practical indicator of lubrication conditions and as a regression target, providing an initial benchmark to show whether sensed data might be informative about lubrication conditions. In this vein, Jakobsen et al. demonstrated that simple time-domain features from a low-cost US microphone can estimate κ [?]. To the authors' knowledge, no corresponding study of κ estimation from AE exists. Moreover, multi-sensor approaches are regarded as a route to more robust bearing monitoring [?], and because AE and US respond to the same contact phenomena through different physical mechanisms and at different frequency scales, simultaneously acquired signals may carry complementary information. This has not been tested, since simultaneous AE-US measurements from the same bearing have not been available.

The present study introduces, to the authors' knowledge, the first simultaneous, time-synchronised acquisition of wideband acoustic emission and passive ultrasound from a single ball bearing under controlled variation of rotational speed and temperature. Using this dataset, the paper makes two principal contributions. First, it demonstrates that fusing features extracted from AE and US signals significantly improves estimation of the ISO 281 viscosity ratio, κ , compared with either sensing modality alone. Second, correlation analysis and SHAP-based interpretation suggest that this improvement arises because AE and US encode partly different aspects of the operating conditions.

First, AE and US data are collected from a bearing test rig under varying operating conditions. Time- and frequency-domain features are extracted from frequency sub-bands of each channel and analysed with respect to κ and the operating conditions. These features are then used for κ regression with linear and non-linear models under an acquisition-grouped evaluation protocol that prevents leakage between training and evaluation data. Finally, SHAP-based interpretation is used to examine the contributions of the two sensing modalities. Because lubricant degradation and contamination are not independently varied, the study addresses acoustic estimation of nominal ISO 281 lubrication adequacy rather than direct detection of starvation, ageing, or contamination.

The remainder of the paper is organised as follows. Section 2 reviews the theory of κ and the physical origins of AE and US signals. Section 3 describes the experimental setup and data collection. Section 4 presents the signal-processing and feature-extraction pipeline, including the characterisation of the frequency sub-bands. Section 5 reports the feature analysis, including the marginal and conditional operating-condition correlations. Section 6 presents the regression models and leakage-controlled evaluation protocol. Section 7 reports the results, Section 8 discusses the findings, and Section 9 concludes.

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2 Background

2.1 Lubrication regimes and the viscosity ratio κ

Ball bearings operate in one of three lubrication regimes depending on the extent to which the lubricant film separates the rolling surfaces. In the full-film elastohydrodynamic lubrication (EHL) regime the surfaces are completely separated and asperity contact is suppressed; in the mixed regime the film is partially formed and intermittent asperity contact occurs; in the boundary regime the film is insufficient and metal-to-metal contact dominates. The ISO 281 viscosity ratio κ provides a continuous, dimensionless indicator of the lubricant viscosity relative to that required for full EHL film formation. It is commonly used for lubrication regime classification with approximate boundaries at $\kappa < 0.1$ (boundary), $0.1 \leq \kappa < 0.4$ (mixed, severe), $0.4 \leq \kappa < 1$ (mixed, moderate), and $\kappa \geq 1$ (full film) [? ?]. The ISO 281 viscosity ratio is defined as

$$\kappa(T, n, d_{pw}) = \frac{\nu(T)}{\nu_1(n, d_{pw})}, \quad (1)$$

where $\nu(T)$ [mm² s⁻¹] is the actual kinematic viscosity of the lubricant at operating temperature T , and $\nu_1(n, d_{pw})$ is the minimum kinematic viscosity required for full EHL at the bearing's rotational speed n [rpm] and pitch diameter d_{pw} [mm] [?]. The reference viscosity is given by

$$\nu_1(n, d_{pw}) = \begin{cases} 45000n^{-0.83}d_{pw}^{-0.5} & n < 1000 \text{ rpm}, \\ 4500n^{-0.5}d_{pw}^{-0.5} & n \geq 1000 \text{ rpm}. \end{cases} \quad (2)$$

The operating viscosity ν is computed from the ASTM D341 Walther viscosity–temperature equation [?] fitted to the lubricant's 40 °C and 100 °C reference values, with a standard value of $\lambda = 0.7$. Since ν_1 decreases with increasing rotational speed, κ captures the combined effect of speed, temperature, and lubricant grade on lubrication adequacy at a given operating point. Importantly, κ assumes nominally intact lubricant and sufficient lubricant supply: the Walther equation reflects the properties of fresh lubricant, so κ does not encode viscosity changes from degradation or contamination, nor does it account for starvation. At high rotational speeds, replenishment may not keep pace with lubricant displaced by the rolling elements, reducing the actual film below κ 's prediction even when the bulk lubricant is intact [?].

2.2 Acoustic emission and passive ultrasound for lubrication monitoring

AE refers to transient elastic stress waves generated by sudden energy releases in a material, where asperity contact and micro-fracture are the dominant sources in rolling bearings [? ?]. Under boundary and mixed lubrication, intermittent asperity contacts generate broadband stress waves that propagate through the bearing structure and are detected by AE sensors coupled to the outer housing [? ?]. The emission rate, amplitude, and frequency content of these events are therefore sensitive to changes in contact severity, with intensity decreasing as film adequacy improves. Transmission path loss is inhomogeneous and path-dependent, affecting both the amplitude and frequency content of the received signal [?]. AE sensors for bearings respond from tens of kHz up to around 1 MHz.

In bearing monitoring, ultrasound has conventionally meant *active*, reflection-based probing such as film-thickness measurement [?]. Passive US sensing instead detects structure-borne acoustic energy emanating from the bearing during operation. Ultrasonic emissions span from 20 kHz into the MHz range, overlapping much of the AE band. The level and spectral character of US signals are sensitive to the lubrication condition, friction level, and asperity contact activity [? ?]. Commercial handheld probes (UE Systems, Sonotec, SDT Ultrasound) typically use hetero-

dynded US down-mixed to the audible range to enable human-assisted diagnostics; the sensors are narrowband, with peak sensitivity at their resonance frequency.

The two sensing methods thus respond to the same underlying contact events—asperity contacts, rapid pressure changes, friction, particle impacts, and crack growth—but through different physical mechanisms and at different frequency scales. AE captures individual high-frequency transient events from discrete micro-fractures and asperity collisions, while passive US reflects structural resonances, friction noise, and the aggregate effect of contact activity. Both channels should therefore carry information about κ , but their different sensitivities to transient events and continuous contact phenomena suggest that AE and passive US may provide complementary information about lubrication-dependent contact dynamics.

These considerations motivate the experimental investigation that follows, which tests whether this suggested complementarity is borne out by κ -regression performance, and whether any gain from combining the channels can be attributed to explainable differences in their information content.

3 Experimental Setup

To test the complementarity of AE and US under controlled conditions, a test rig was designed and built in which rotational speed and temperature can be varied while load and lubricant are held fixed. AE and US sensors collect time-synchronised data across these operating conditions, and the resulting dataset is used to estimate κ . This section presents the rig setup, the experimental protocol for varying operating conditions, the data acquisition method, and an overview of the acquired data.

3.1 Experimental rig

The test rig is a custom-built bearing test stand at Aarhus University (Herning, Denmark). It can be seen in Fig. 1.

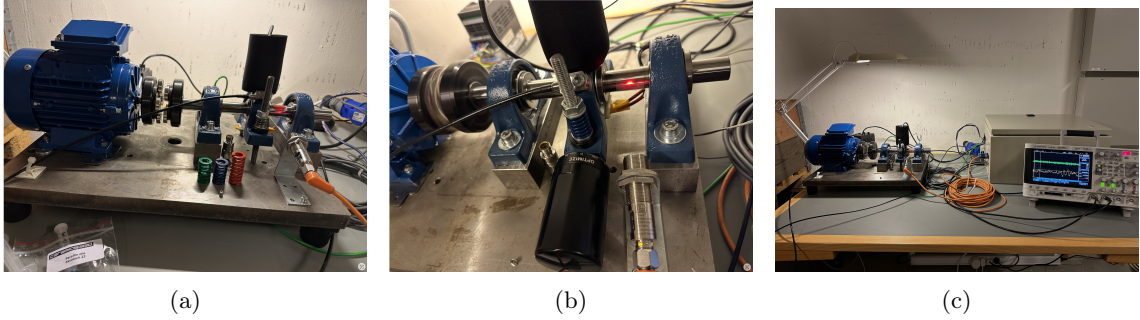


Figure 1: The custom-built bearing test rig at Aarhus University (Herning, Denmark).

The rig uses a deep-groove ball bearing of model SYJ25 with 25 mm bore diameter, pitch diameter $d_{pw} = 38.0$ mm, 9 steel rolling elements, and dynamic load rating 14000 N. The bearings are mounted using pillow blocks to allow mounting sensors as close as possible to the source of the signals. The bearing is driven by a Klee T712-2 0.55 kW variable-frequency electric motor through a variable-frequency drive (VFD) and a PID controller regulates the speed based on rotational-speed measurements from an OGT5000 diffuse-reflection sensor. Temperature is increased by a heating element mounted on the bearing pillow block, regulated by an Omron E5CC PID controller using a type K sensor mounted on the outer ring of the bearing. A spring load applies a nominal radial load of 60 kg and the load measured by an FN2114 pedal load cell is approximately 61 kg. The bearing is lubricated with 0.5 ml of Keratech 22 oil (kinematic viscosity $\nu_{40} = 22$ cSt, $\nu_{100} = 4.1$ cSt).

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AE signals are collected by a Kistler Piezotron Type 8152C1 wideband piezoelectric sensor bonded to the bearing housing. Its output is conditioned by a Kistler Type 5125C AE Piezotron coupler, which applies a Butterworth band-pass filter from 50 kHz to 1 MHz to the raw AE signal before it is digitised. This band-pass sets the 1 MHz upper analysis limit used throughout the paper.

How is it mounted?

US signals are collected by a UE Systems Ultra-Trak 750 sensor mounted on the bearing housing. The sensor is narrowband, with peak sensitivity around its resonance frequency. It performs internal heterodyning whereby the frequencies of interest are shifted into the hearable domain from 1-15 kHz, as is typical of commercial US instruments. It is this heterodyned output, rather than the raw acoustic signal, that constitutes the US waveform analysed in this paper, and the US sub-band frequencies referred to in later sections are therefore frequencies of the heterodyned signal.

State the resonance frequency and heterodyne specification from the Ultra-Trak 750 datasheet.

Both channels are digitised by the same Agilent MSO2024 oscilloscope at 12.5 MHz, sharing a sampling rate and sampling points, so every acquired segment contains inherently time-synchronised AE and US waveforms, with no post-hoc alignment required.

3.2 Experimental protocol for varying κ

The experiment follows a protocol which varies the nominal operating conditions through changes in rotational speed (RPM) and housing temperature ($^{\circ}\text{C}$). This causes the viscosity ratio κ of the bearing in the test rig to vary under controlled loading and bearing geometry. The nominal rotational speed follows a staircase sweep from 100 to 3000 rpm and back down to 100 rpm in steps of 100 rpm with a minimum 60-second hold at each step. This staircase is repeated for 13 temperature blocks. The nominal temperature is increased in coherent blocks from 40 to 100 $^{\circ}\text{C}$ in 5 $^{\circ}\text{C}$ increments. After each downsweep from 3000 rpm to 100 rpm, the housing is brought to the next temperature set-point before the next staircase sweep begins. The hold at the 3000 rpm staircase peak is substantially longer than the ordinary 60-second steps (approximately 20 minutes), so this operating point is over-represented in the data: it accounts for roughly a third of all sweeps. The total duration of the experiment is approximately 14 hours.

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Over the ≈ 14 -hour experiment the lubricant is assumed to remain nominally intact: no degradation, starvation, or contamination is expected, so the lubrication conditions vary only through rotational speed and temperature and their effect on the lubricant, consistent with the assumptions underlying κ .

3.3 Acquisition segments and measured operating conditions

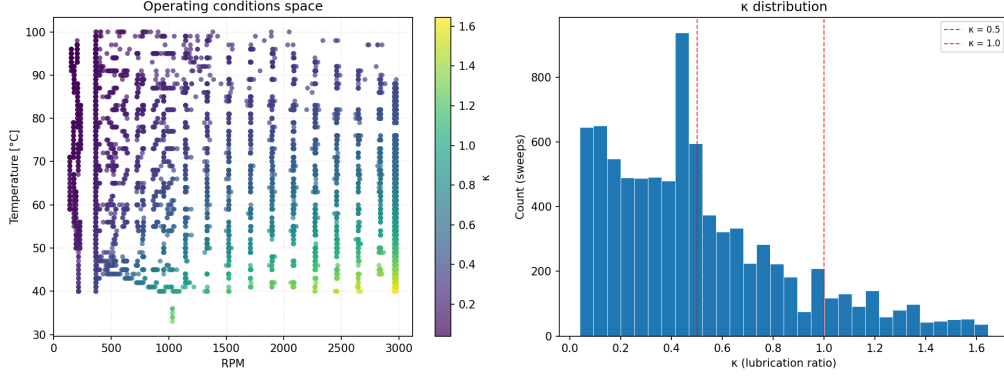
Data are acquired as short segments rather than continuously: every ~ 5.9 s the oscilloscope captures a 20 ms segment ($N \approx 2.5 \times 10^5$ samples per channel) containing time-synchronous AE and US waveforms, together with the rotational speed and housing temperature at the segment sampling time. Throughout this paper, one such segment serves as the unit of observation for all feature extraction, model training, and evaluation.

Include figure which illustrates the protocol for varying operating conditions and kappa once the speed issue has been settled.

Since the operating conditions are maintained through a control loop, they vary from the nominal conditions specified by the protocol. The measured operating conditions span 140–2977 rpm and 33–100 $^{\circ}\text{C}$; the measured temperature range extends below the commanded 40 $^{\circ}\text{C}$ floor (down to $\approx 33^{\circ}\text{C}$), which is tentatively attributed to instability of the temperature control rather than to any commanded set-point. During the last temperature block, the speed recordings at the top of the speed staircase were corrupted and discarded.

After excluding an initial stationary warm-up segment recorded before the first staircase and 235 segments with measured rotational speed above 3000 rpm, 8 504 segments per channel remain

(median 10 segments per staircase hold). The measured operating-condition space and the distribution of calculated κ values for the remaining segments are shown in Fig. 2, and their evolution over the course of the experiment in Fig. 3. The κ value for each segment is computed according to Equations (1) and (2) using the per-segment measured rotational speed and temperature, with ν evaluated via the ASTM D341 Walther equation. The κ values range from full film ($\kappa \approx 1.6$) to boundary lubrication ($\kappa \approx 0.05$).



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Figure 2: Left: scatterplot of the measured operating conditions (rotational speed and temperature) during the experiment. Each point corresponds to a 20 ms segment of sampling at a given rotational speed and temperature. The colour of each point corresponds to the derived κ value for that segment. Right: distribution of the derived κ values.

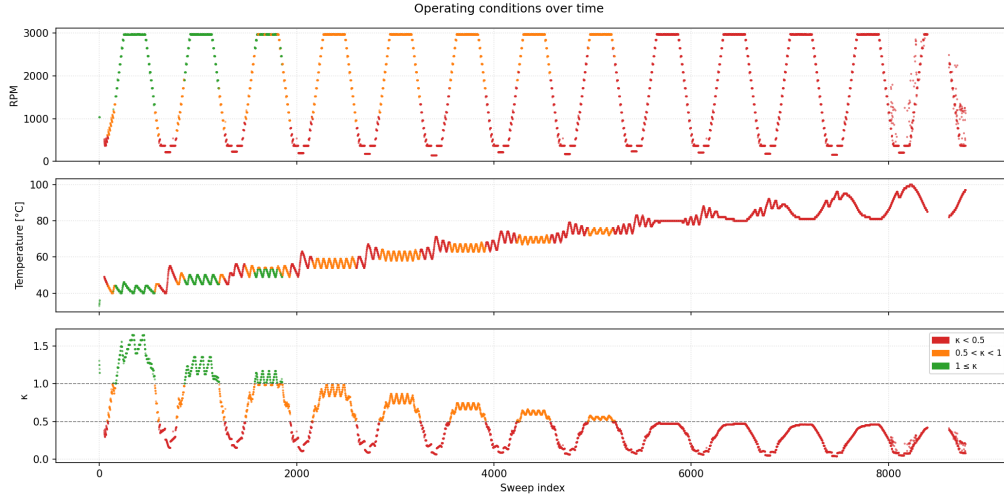


Figure 3: Measured operating conditions (rotational speed and temperature) during the experiment, ordered chronologically by acquisition time. Each point corresponds to a 20 ms segment of sampling. The colour of each point corresponds to the derived κ value for that segment.

The experiment thus yields 8 504 time-synchronised AE–US segment pairs from a single bearing, with per-segment measured operating conditions and κ spanning the boundary to full-film regimes ($\kappa \approx 0.05$ –1.6). Because both channels observe the same contact events at the same instants, differences in their information content are attributable to the sensing methods themselves — the property on which the complementarity test rests. Section 4 next describes the signal processing that converts these segments into features for the κ regression.

4 Signal Processing and Feature Extraction

This section converts the acquired segments into the feature sets used for regression: the signals are first inspected qualitatively, then decomposed into frequency sub-bands, and finally summarised by a set of time- and frequency-domain features per band.

4.1 Waveforms and spectrograms

Fig. 4 shows the waveforms of four segments for both the AE and US channels, and Fig. 5 shows the spectrograms of all segments over time. AE and US qualitatively display clearly different waveforms and frequency content in all segments. AE in general contains spectral energy from 50 kHz to 1 MHz as expected from the AE sensor and band-pass filter applied during data acquisition. Some energy also appears above 1 MHz, but as its origin could not be reliably established it is excluded from subsequent analysis. US shows highest energy concentration in the spectral band from 0 kHz to 20 kHz, consistent with heterodyning, but also displays energy from 20 kHz to 100 kHz. This is attributed to heterodyning artifacts consisting of sums and multiples of the carrier and original frequencies.

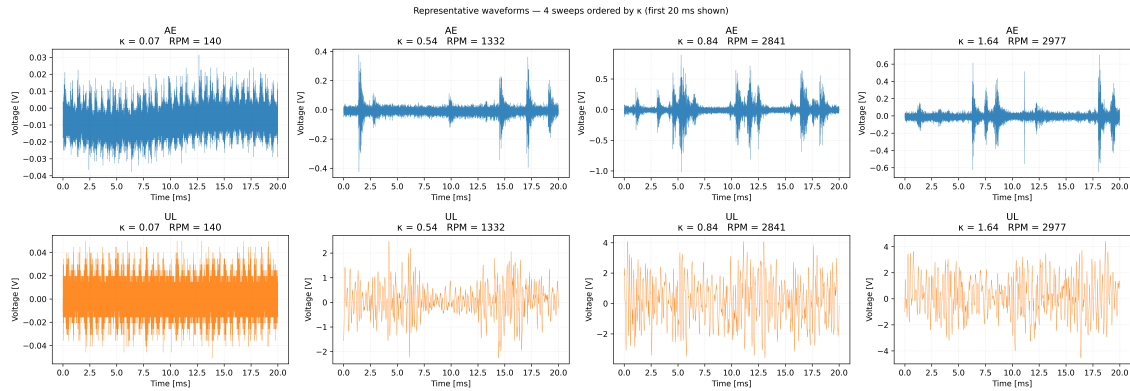


Figure 4: Waveforms of AE and US signals from the same segments for different values of κ . AE shows transient energy bursts consistent with impulsive contact-related events. US shows a more regular, lower frequency, non-stationary waveform.

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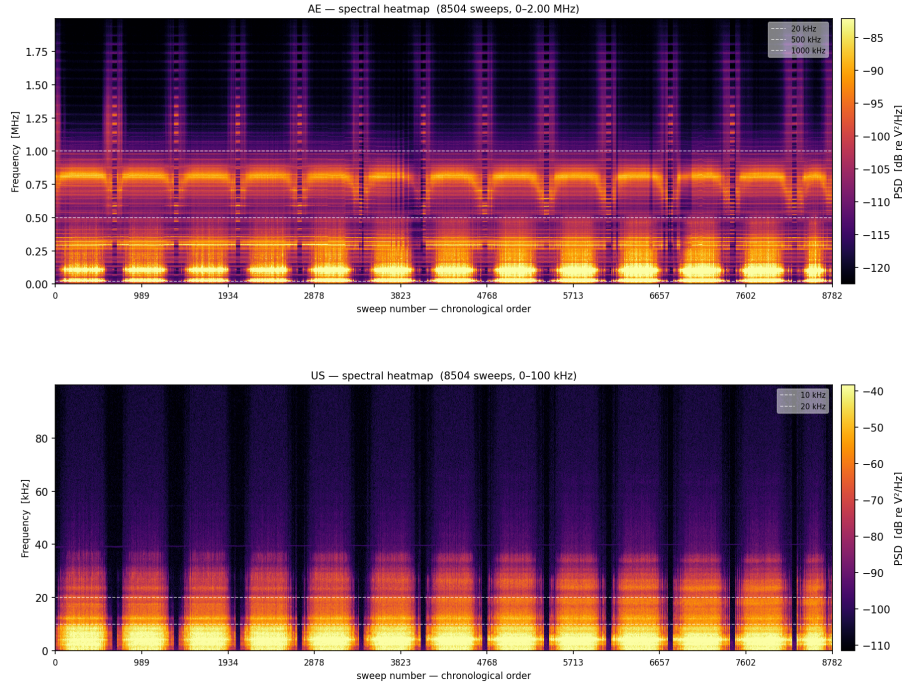


Figure 5: Spectrograms of the raw AE and US ordered chronologically by acquisition time. PSD is calculated with Welch’s method, Hann windows, 50% overlap, and segment lengths of 4096 samples (AE) and 65536 samples (US).

4.2 Sub-band decomposition

The AE and US signals are decomposed into frequency sub-bands prior to feature extraction, since the predictive information related to κ is expected to be more localised in sub-bands than in broadband signals [? ? ? ?]. For the AE channel, two sub-bands are extracted in addition to the broadband signal: 20–500 kHz and 500 kHz–1 MHz, containing bands of spectral energy which upon visual inspection of Fig. 5 vary with the operating conditions. Content above the 1 MHz coupler low-pass is excluded from the feature set. The coupler additionally high-passes at 50 kHz before digitisation (Section 3.1), so the AE broadband spans 50 kHz–1 MHz and the 20–500 kHz sub-band—a nominal digital passband—carries effective content of 50–500 kHz, as no energy remains below 50 kHz. For the US channel, the signal spans the probe’s 0–100 kHz acquisition band, and three sub-bands are extracted in addition to the broadband signal: 0–10 kHz and 10–20 kHz within the nominal heterodyned baseband, and 20–100 kHz. The 0–20 kHz bands correspond to the audible heterodyned baseband commonly used in commercial passive ultrasound instruments, while 20–100 kHz evaluates whether additional ultrasonic information retained after acquisition contributes to prediction. The sub-bands are summarised in Table 1. The goal of sub-band feature extraction is not to find the most informative, physically motivated frequency bands, but to extract more information than from simple broadband features. Usefulness of sub-bands is expected to vary across test rig setups, bearings, lubricants, and operating conditions, so data mining for optimal sub-bands is furthermore expected to overfit on this particular setup. All decompositions use 5th-order Butterworth filters applied forward–backward giving zero phase distortion and a 10th-order effective magnitude response.

Finalize spectrograms with correct colors, ranges, and dashed lines for sub-bands

Channel	Sub-band	Filter type	Frequency range
AE	Broadband	Band-pass	50 kHz–1 MHz
AE	Sub-band 1	Band-pass	20–500 kHz
AE	Sub-band 2	Band-pass	500 kHz–1 MHz
US	Broadband	Low-pass	0–100 kHz
US	Sub-band 1	Band-pass	0–10 kHz
US	Sub-band 2	Band-pass	10–20 kHz
US	Sub-band 3	Band-pass	20–100 kHz

Table 1: Frequency sub-bands used for signal pre-filtering before feature extraction. AE ranges reflect the coupler’s 50 kHz–1 MHz passband (Section 3.1); the 20–500 kHz sub-band is the nominal digital filter, with effective content 50–500 kHz.

4.3 Feature set

Features are computed from the voltage signal of each segment–sensor pair, for the broadband signal and for each sub-band. In total 14 features are computed, comprising 8 time-domain and 6 frequency-domain statistics, as listed in Table 2. The set is deliberately canonical: commonly used features such as standard deviation, variance, peak, impulse factor, and RMS frequency are approximately equivalent to products of the chosen features and are therefore omitted, so that no feature is a deterministic function of another. All features (without the specific sub-bands) have previously been applied to signals for bearing condition monitoring and LCM [? ?]; the Hjorth parameters originate from EEG analysis [?]. Mobility, the ratio between the standard deviations of a signal’s first time-derivative and of the signal itself, is a time-domain estimate of the signal’s mean frequency: it rises as energy shifts toward higher frequencies within the band. Complexity, the mobility of the derivative relative to the mobility of the signal, measures spectral spread: it attains its minimum of one for a pure sinusoid and grows as the signal departs from a single dominant frequency toward broadband content.

Name	Formula
<i>Time domain</i>	
RMS	$\sqrt{N^{-1} \sum_n x_n^2}$
Skewness	$N^{-1} \sum_n [(x_n - \bar{x})/\sigma]^3$
Kurtosis	$N^{-1} \sum_n [(x_n - \bar{x})/\sigma]^4$
Crest factor	$x_{\text{peak}}/x_{\text{RMS}}$
Shape factor	$x_{\text{RMS}} / (N^{-1} \sum_n x_n)$
Margin factor	$x_{\text{peak}} / (N^{-1} \sum_n \sqrt{ x_n })^2$
Hjorth mobility	$M(x) = [\text{Var}(x')/\text{Var}(x)]^{1/2}$
Hjorth complexity	$M(x')/M(x)$
<i>Frequency domain</i>	
Dominant frequency	$f_{\arg \max_k S_k^2}$
Centre frequency	$f_c = \sum_k f_k S_k / \Sigma S$
Spectral bandwidth	$\sigma_w = (\sum_k (f_k - f_c)^2 S_k / \Sigma S)^{1/2}$
Spectral skewness	$\sum_k (f_k - f_c)^3 S_k / (\sigma_w^3 \Sigma S)$
Spectral kurtosis	$\sum_k (f_k - f_c)^4 S_k / (\sigma_w^4 \Sigma S)$
Spectral flatness	$\exp(K'^{-1} \sum_{k: S_k > 0} \ln S_k) / \bar{S}$

Table 2: The 14 extracted features: 8 time-domain and 6 frequency-domain. Notation: N signal length; $n, k = 1, \dots, N$; $\bar{x}$ sample mean; σ population standard deviation; $x_{\text{peak}} = (\max x - \min x)/2$ half the peak-to-peak range; x' the first time-derivative of the signal; $S_k = |X_k|$ one-sided FFT magnitude at bin k ; f_k the corresponding frequency; $\bar{S}$ arithmetic mean of $\{S_k\}$; $\Sigma S = \sum_k S_k$; K' the number of bins with $S_k > 0$.

All frequency-domain features are computed from the one-sided FFT magnitude spectrum of

the segment. The spectral skewness and kurtosis are standardised by $\sigma_w^p \Sigma S$ and are therefore dimensionless and invariant to overall signal amplitude, so that spectral-shape information is separated from the amplitude information carried by RMS.

Extracting the 14-feature set independently for the broadband signal and each sub-band yields 42 AE features (broadband plus two sub-bands) and 56 US features (broadband plus three sub-bands) per segment. Features are named by channel, band, and statistic (e.g. AE_500-1000kHz__mobility). broadband features carry no band prefix. These two sets, and their union (the combined AE+US set, 42 + 56 features), define the three sensor configurations—AE-only, US-only, and combined.

The extracted features are not assumed to represent lubrication-specific quantities. Instead, they provide a compact representation of acoustic signal characteristics that may contain contributions from lubrication-dependent contact dynamics, operating conditions, structural transmission effects, and sensor coupling. The subsequent analysis therefore examines what the features respond to (Section 5) and whether the two channels contribute complementary information to κ regression (Sections 6–7).

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5 Feature Analysis

This section first establishes that the 42 AE and 56 US features are informative about κ . It then examines their operating-condition dependence, distinguishing marginal from conditional correlations, which reveals potentially complementary information for inferring κ .

5.1 Correlation with κ

Feature- κ correlation is quantified by Spearman’s rank correlation coefficient ρ , computed at segment level. Both channels carry substantial information about κ , as shown in Fig. 6. The strongest AE feature, the 500 kHz–1 MHz Hjorth mobility, correlates with κ at $|\rho| = 0.85$, and the strongest US feature, the 10–20 kHz RMS, at $|\rho| = 0.69$; in total, 23 of the 42 AE features and 10 of the 56 US features exceed $|\rho| = 0.4$. Correlations of this strength indicate that κ regression from either channel alone is feasible.

Wire the feature- κ correlation numbers (and the other hand-typed ρ values in this section) as macros from a new-pipeline export step.

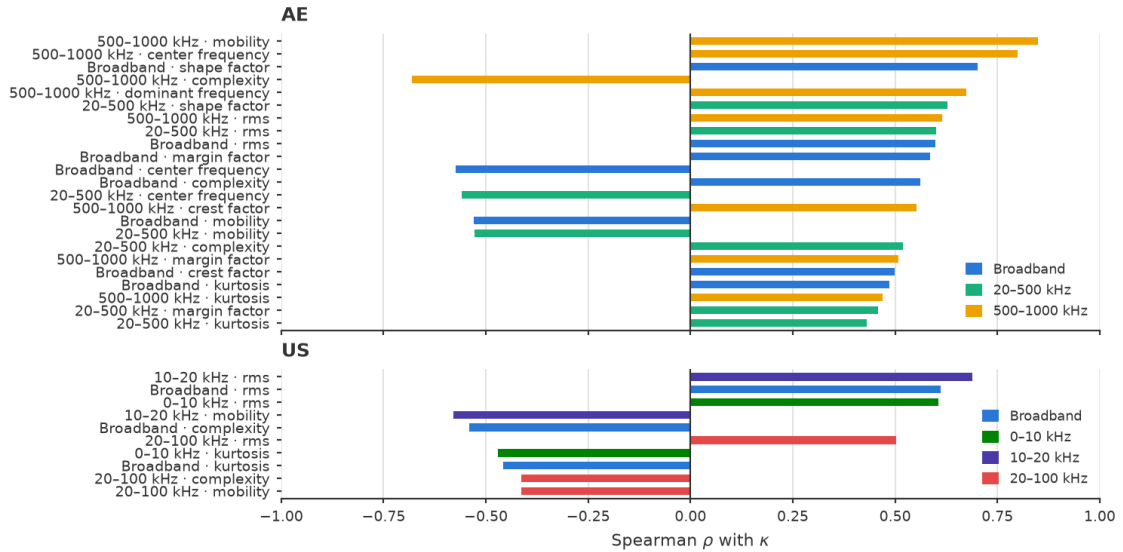


Figure 6: Signed Spearman correlation with κ for all candidate features with $|\rho| \geq 0.4$, sorted by absolute correlation within channel: AE (top) and US (bottom). Colours denote the frequency sub-band of each feature; features below the threshold are omitted.

Correlation with κ , however, reveals nothing about complementarity. Because κ is a determin-

istic function of rotational speed and temperature (Eq. (1)), two features may correlate equally strongly with κ while responding to entirely different components of the operating point or to exactly the same one. Whether AE and US are complementary is therefore a question of which operating-condition dependencies each channel exhibits, examined in the remainder of this section.

5.2 Marginal and conditional correlation with operating conditions

The marginal correlation between a feature and an operating condition (speed or temperature) is the Spearman correlation computed over all segments, with the other operating variable left free to vary. The marginal correlation of all features with both speed and temperature can be seen in the left panel of Fig. 7. Both channels correlate with κ through their dependence on speed and temperature, but speed is the dominant axis for both, with energy features such as RMS positively correlated with speed. The correlations with speed are, however, predominantly positive for AE and negative for US. A notable case is the US mobility feature identified as most important by Jakobsen et al. [?], which correlates only weakly with speed and more strongly with temperature. Interestingly, mobility in the 20–500 kHz and 500 kHz–1 MHz AE bands is negatively and positively correlated with speed, respectively. The marginal view is in general dominated by the speed sweep, however. We therefore condition on each operating variable in turn.

The right panel of Fig. 7 shows the partition-based conditional Spearman correlations, in contrast to the marginal correlations in the left panel. In the conditional analysis, segments are grouped either into 100 rpm staircase steps, giving the within-step correlation with temperature at held speed, or into 2°C temperature bins, giving the within-bin correlation with speed at held temperature. A partition is retained only if it holds at least 30 segments and spans at least 8 distinct levels of the variable being correlated against, so that each within-partition correlation is reliable. The temperature axis is then summarised over 22 valid rpm-step partitions (median 259 segments and 39 distinct temperatures each) and the speed axis over 31 valid temperature bins (median 232 segments and 17 distinct speed steps each). The median correlation across partitions is reported.

Conditioning reveals a clear separation between the two channels. The speed axis is essentially unchanged: the strongest AE feature (500 kHz–1 MHz mobility, $\rho = 0.89$) and US feature (0–10 kHz RMS, $\rho = 0.91$) retain near-marginal conditional correlations with speed at fixed temperature, confirming that both channels genuinely track rotational speed. The temperature axis, however, diverges: AE features show moderate conditional temperature correlations (strongest: 20–500 kHz mobility, $\rho = -0.56$), whereas US features correlate with temperature substantially more strongly and with mixed sign across sub-bands (0–10 kHz mobility $\rho = -0.79$; 10–20 kHz mobility $\rho = +0.78$). AE and US therefore occupy partly distinct regions of the conditional correlation plane. Because κ depends on both speed and temperature (Eq. (1)), the two channels encode the operating point through partly distinct operating-condition channels: AE primarily through speed, US through both speed and temperature. These observations suggest that AE and US encode partly different components of the operating point and hence complementary information for κ regression; whether this translates into improved performance when the channels are combined is tested in Section 6.

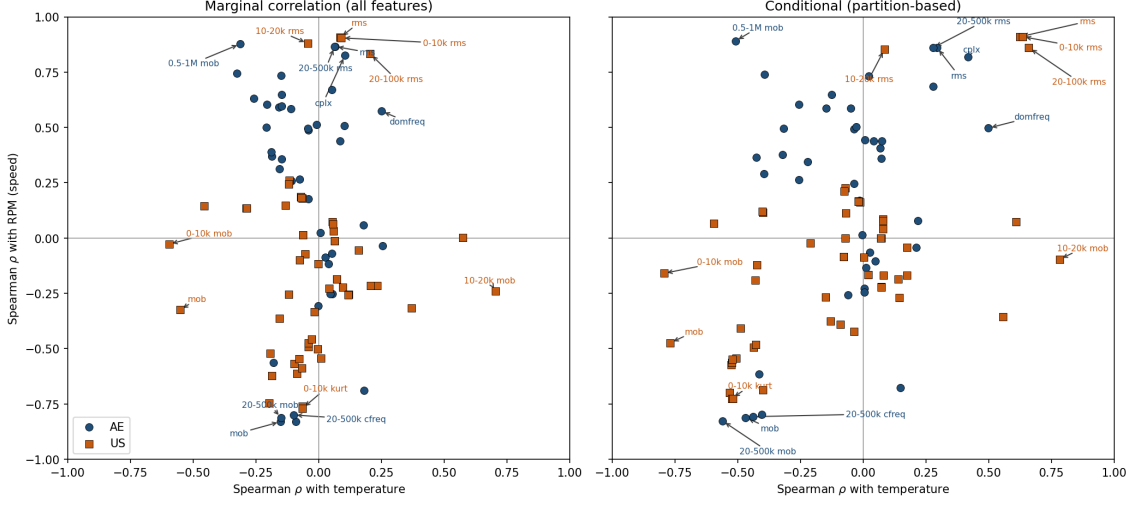


Figure 7: Marginal versus conditional Spearman correlation structure of the AE (circles) and US (squares) candidate features. Each point is one feature and axes are the Spearman ρ with measured temperature and rotational speed. *Left*: marginal correlations over all segments. *Right*: partition-based conditional correlations—within-speed-bin for the temperature axis and within-temperature-bin for the speed axis. Labels and arrows identify the five features per sensor with the largest absolute conditional correlation on each axis. The speed axis is essentially unchanged by conditioning: both channels seem to track rotational speed. The temperature axis spreads substantially on conditioning and US features seem to correlate with temperature more strongly than AE and with mixed sign across sub-bands.

6 Regression Modelling and Statistical Evaluation

This section uses the features extracted in Section 4 for κ regression through two different model families. The goal is to assess whether κ regression is feasible with the extracted features and whether any complementarity between the AE and US features sets improve κ regression when combine. The section describes the experimental design, including data splits and fitting protocol, the model families, and the statistical tests used to evaluate the significance of the results; the results themselves are reported in Section 7.

6.1 Experimental design

Three feature sets are evaluated: AE-only, US-only, and Combined (AE+US). Both model families are fitted to each, giving six model–sensor configurations in total. Models are trained on the full candidate feature sets (42 AE and 56 US features) and no feature selection is applied, as preliminary experiments showed no accuracy benefit from it. Interpretation is instead performed post hoc with redundancy-aware clustered SHAP.

The segments from which features are calculated are grouped to prevent soft leakage of information: segments collected close together in time or under matching operating conditions encode near-duplicate signal characteristics and must not straddle a fold boundary. The grouping protocol consists of two steps:

1. All segments recorded during the same 60s operating-condition hold are grouped into one acquisition-hold group.
2. Because the staircase protocol revisits each operating point on the up- and down-sweep, and κ is deterministic in the operating point, hold groups whose mean measured rotational speed and mean measured temperature fall within the same bin ($100 \text{ rpm} \times 1^\circ\text{C}$) are merged into

a single group (twins). Without this step, a model could score on a held-out operating point through its trained twin.

This yields 520 groups of 8498 segments in total.

The groups are evaluated by a single-pass 5-fold grouped cross-validation over the full dataset [? ? ?]. Each fold is held out exactly once, while hyperparameter selection and model fitting are repeated from scratch on the remaining four folds, following the nested cross-validation principle [? ? ?]. Pooling the five held-out folds yields one prediction for every segment from a model that never saw its group during training or tuning. All reported performance figures and significance tests are computed on these pooled held-out predictions. Performance is quantified by the coefficient of determination (R^2), the root-mean-square error (RMSE), and the mean absolute error (MAE). The mean per-fold grouped-CV R^2 is reported alongside as an indicator of fold-to-fold stability. RMSE is used for the significance tests (Section 6.3).

6.2 Model families

Two regression model families, summarised in Table 3, are evaluated across the three sensor configurations – a linear model (Elastic Net) and a non-linear model (LightGBM). The families are chosen for their widespread use, their regularisation abilities, and their contrast to each other, which may highlight whether estimation of κ benefits from using a non-linear model instead of a linear one. The goal is not to find the best performing model among a large corpus, but to uncover general trends in κ estimation, whether complementarity of AE and US signals exists for either model family, and whether complementarity is affected by model family.

Both models are tuned by a group-aware inner cross-validation within each outer training fold so that fold scores are never contaminated by hyperparameter selection on the held-out data. Elastic Net and LightGBM are tuned with `ElasticNetCV` and `Optuna`, respectively. For Elastic Net, features are standardised to zero mean and unit variance inside the per-fold pipeline, so the scaling parameters are learned on the training folds only and never leak across the split. LightGBM operates on the raw features, as tree splits are invariant to monotone feature scaling. The inner folds are derived from the outer fold partition, so the same twin-merged grouping governs both levels of the nested procedure.

Model	Rationale	Hyperparameters & tuning
Elastic Net	$\ell_1 + \ell_2$ regularised linear model; ℓ_1 promotes sparsity, ℓ_2 stabilises correlated features.	α (9 values, 10^{-4} – 10^2) $\times$ ℓ_1 ratio (6 values, 0.1–1.0); <code>ElasticNetCV</code> , inner 4-fold grouped.
LightGBM	Gradient-boosted trees; captures non-linear feature– κ mappings and high-order interactions.	Learning rate, num. leaves, max. depth, min. child samples, ℓ_2 reg.; Bayesian optimisation (<code>Optuna</code> TPE, 20 trials per outer fold).

Table 3: Regression model families: rationale and hyperparameter tuning strategy.

6.3 Statistical significance testing

Three cross-sensor comparisons form the study’s focal claims: AE vs. US, Combined vs. AE, and Combined vs. US, each within both model families. All configurations are scored on the identical pooled held-out segments, so every comparison is fully paired: for configurations A and B , each segment i yields a squared-error differential $d_i = e_{B,i}^2 - e_{A,i}^2$, so that positive values favour the first-listed configuration A . The differentials are averaged within each holdout group, so that overly similar segments do not overstate the significance, giving 520 group-mean differentials. These are tested against the null hypothesis of a zero-mean differential with a two-sided Diebold–Mariano

test [?] using the Harvey et al. small-sample correction [?], at a significance level of $\alpha = 0.05$. A naive segment-level test on the 8498 near-duplicate segments would overstate this certainty by orders of magnitude; only the grouped test is reported. No multiple-comparison correction is applied; all six contrasts remain significant under a Bonferroni correction across the six tests.

7 Results

This section presents the results from the pipeline described in the paper until now. This includes overall performance with visualisation of the κ regression, the performance measures, the sensor comparison and fusion, and the results of the statistical tests.

7.1 Overall performance

The pooled held-out predictions of all six configurations are shown in Fig. 8. The figure shows predicted vs. actual κ for both model families and all sensor configurations, coloured by rotational speed. Across the grid, predictions track the diagonal most closely for LightGBM whereas Elastic Net compresses the range, over-estimating at low-to-mid κ -levels and under-estimating at high κ -levels. Predictions fall into discrete vertical stripes, one per operating point visited by the staircase. All segments at a given operating point share the same κ , so each stripe’s vertical scatter is the model’s within-operating-point estimation noise. This scatter is largest toward the sparsely sampled full-film range ($\kappa > 1$).

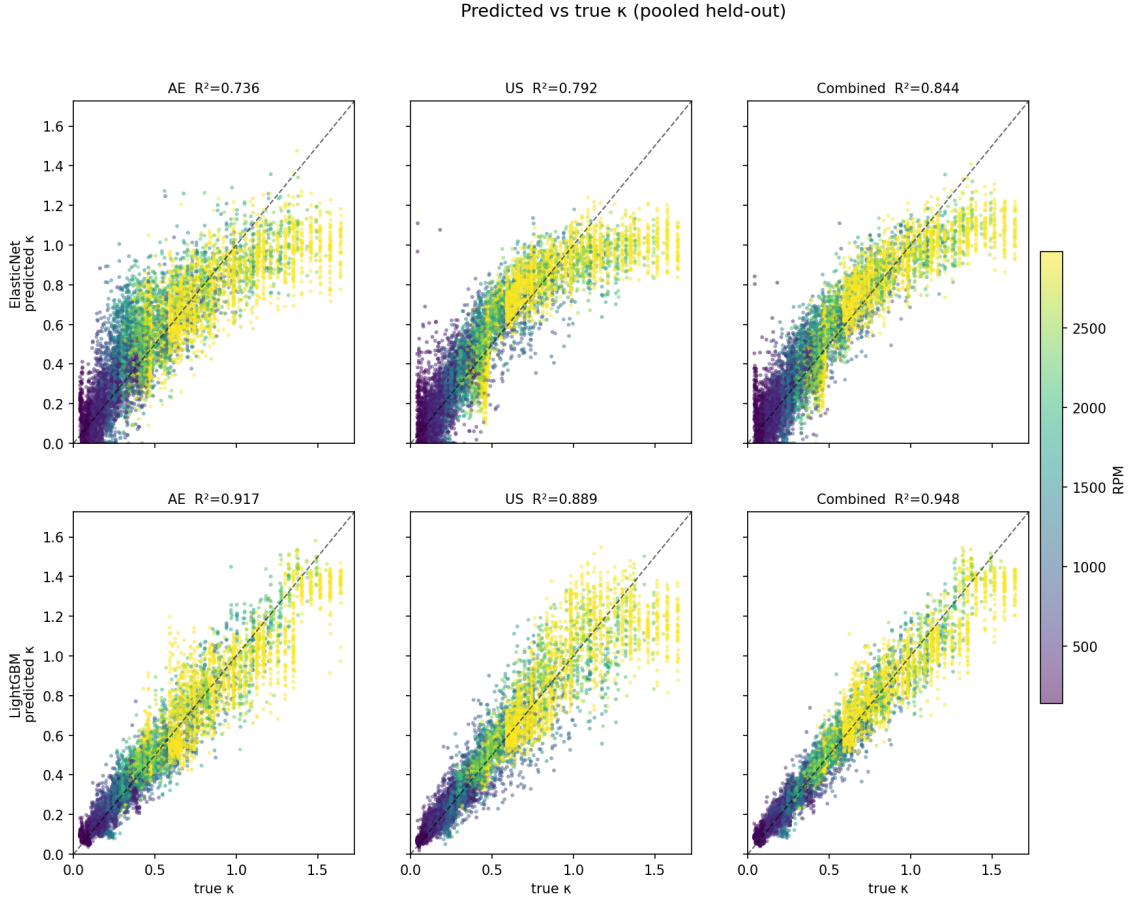


Figure 8: Predicted vs. actual κ (pooled held-out predictions) for both model families and all sensor configurations, coloured by rotational speed. The dashed diagonal indicates perfect prediction. Residual spread is largest toward the sparsely sampled full-film range ($\kappa > 1$).

Table 4 quantifies these impressions: all six model–sensor configurations predict κ well, with

LightGBM ahead of Elastic Net in every one. The size and consistency of this gap indicate a substantially non-linear feature- κ mapping, and the close agreement between the per-fold and pooled held-out scores indicates stable generalisation. Whether the two channels achieve this accuracy through the same or through different information is the question taken up next.

Sensor	Model	CV R^2	R^2	RMSE	MAE
AE	Elastic Net	0.727	0.737	0.183	0.135
	LightGBM	0.905	0.917	0.103	0.071
US	Elastic Net	0.787	0.792	0.162	0.120
	LightGBM	0.882	0.889	0.119	0.076
Combined	Elastic Net	0.839	0.844	0.141	0.105
	LightGBM	0.943	0.948	0.081	0.055

Table 4: Regression model performance: mean grouped-CV R^2 per fold and pooled held-out R^2 , MAE, and RMSE. Best pooled R^2 in bold.

7.2 Sensor comparison

Which single channel performs better depends on the model family: under LightGBM, AE outperforms US, while under Elastic Net the ordering reverses, and both orderings are significant as seen in Table 5. Table 4 indicates why: Elastic Net recovers a substantially larger share of the US channel’s LightGBM performance (R^2 0.79 of 0.89) than of AE’s (0.74 of 0.92), so the AE feature- κ mapping is markedly less accessible to a linear model. This is consistent with the channel separation of Section 5, where AE estimates κ primarily through rotational speed, whose influence on κ is piecewise (Eq. (2) changes exponent at 1000 rpm). This is a non-linearity which a linear model might not be able to represent. The weaker LightGBM performance of US is also visible in Fig. 8 as wider within-stripe scatter. The better temperature resolution of the US features described in Section 5 might also influence the model families’ ability to recover κ such that a linear model performs better.

7.3 Sensor fusion

The combined feature set outperforms both single channels for both model families, and all four combined-vs-single contrasts are significant under the group-level DM test of Table 5. The absolute R^2 gain is modest, but its consistency across model families and contrasts indicates that each channel contributes non-redundant information.

Comparison	Model	Δ RMSE	DM	p
AE vs. US	Elastic Net	−0.021	−5.52	< 0.001
	LightGBM	+0.016	+4.31	< 0.001
Combined vs. AE	Elastic Net	+0.042	+11.40	< 0.001
	LightGBM	+0.022	+7.16	< 0.001
Combined vs. US	Elastic Net	+0.022	+5.21	< 0.001
	LightGBM	+0.038	+7.70	< 0.001

Table 5: Cross-sensor significance tests for both model families on the pooled held-out predictions. Δ RMSE is the average difference in RMSE, and DM is the Harvey-corrected Diebold–Mariano statistic on the group-averaged squared-error differentials (520 groups). For both, positive = first-listed sensor/set better. p is the corresponding two-sided p -value.

7.4 Clustered SHAP feature importance

SHAP (SHapley Additive exPlanations) values [?] are commonly used in machine learning to assign importance to features and interpret their influence on a model’s predictions. They are used here to identify which AE and US features drive the single-channel regressions, and whether features from both channels rank highly in the combined regression.

The full feature sets in this experiment contain many strongly correlated features, and SHAP importance is split among such near-duplicates, obscuring it. Features are therefore here grouped into correlation clusters. Agglomerative hierarchical clustering with average linkage is applied to their pairwise Spearman correlations (feature to feature), and the dendrogram is cut so that features correlated at $|\rho| \geq 0.8^2$ fall in the same cluster. Each feature is assigned to exactly one cluster. SHAP values are computed with TreeExplainer [?] from a single LightGBM model refit on the full dataset, so the importances are descriptive of the fitted feature- κ mapping and carry no held-out performance claim. Fig. 9 reports each cluster’s importance as the mean absolute per-segment sum of its members’ SHAP values, following the grouped-Shapley approach of Jullum et al. [?]; see [?] for a general treatment of grouped feature importance.

The two channels lean on visibly different clusters. AE importance is dominated by a single high-frequency cluster containing 500 kHz–1 MHz mobility and centre-frequency, roughly a factor of seven above the next cluster. US importance is led by an amplitude (RMS) cluster together with a few distinct features such as 10–20 kHz mobility. The conditional correlations of Fig. 7 tell the same story, with the dominant AE cluster tracking speed and the leading US features leaning towards temperature. The two analyses therefore agree by independent routes.

The clustered SHAP values of the combined feature set show that high-ranking features come from both channels: the AE 500 kHz–1 MHz mobility cluster leads, followed by the US 0–10 kHz and 10–20 kHz mobility features. Notably, the US channel’s own leading cluster in isolation (10–20 kHz RMS) all but disappears from the combined ranking, while its temperature-leaning mobility features persist. This is consistent with their strong conditional temperature correlations see in Fig. 7. The combined model thus retains US features for the temperature information that AE correlates with only weakly, and discards US amplitude features whose speed information is redundant with the AE features the model prefers. This mirrors, at the level of individual features, the complementarity established by the fusion results.

²The leading clusters are insensitive to this threshold, with cuts at 0.7 and 0.9 yielding the same top clusters per channel.

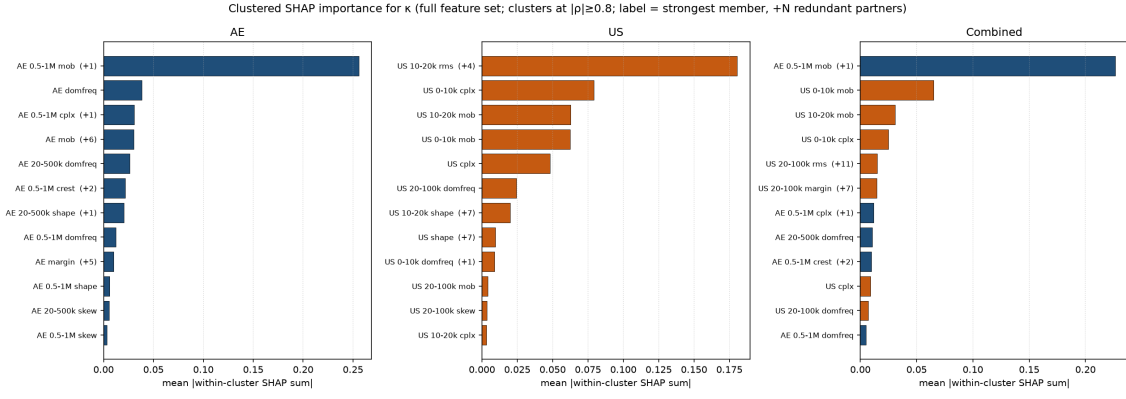


Figure 9: Clustered SHAP importance for κ (LightGBM) on the AE, US, and combined feature sets. Features are grouped into correlation clusters ($|\rho| \geq 0.8$) and each cluster’s mean absolute per-segment SHAP sum is plotted, labelled by its strongest member (with the number of additional correlated members in parentheses); bars are coloured by channel (blue AE, orange US). AE importance concentrates in a 500 kHz–1 MHz mobility cluster and US importance in a 10–20 kHz amplitude (RMS) cluster, mirroring the operating-condition channel separation of Fig. 7. US sub-band labels refer to the probe’s heterodyned audio baseband, not raw acoustic frequencies.

8 Discussion

The principal finding of this work is that combining AE and US features significantly improves κ estimation under a leakage-controlled evaluation protocol. The remaining analyses suggest an explanation for this improvement: AE and US encode partly different aspects of the operating point, particularly with respect to temperature. The subsections below discuss these findings and the limitations of this paper.

8.1 AE–US complementarity and channel encoding

The fusion result is the central finding of the study: the combined feature set outperforms both single channels, for both model families, in all four group-level DM contrasts (Table 5). The gain is modest in magnitude but consistent, and it can be related to the operating-condition encoding of the two channels. Conditionally on speed, US correlates with temperature substantially better than AE (Section 5.2), whereas both channels track speed well. The clustered SHAP results mirror this separation at the feature level, with the combined model retaining temperature-leaning US features while discarding US amplitude features whose speed information the AE features already provide (Section 7.4). The two channels are complementary and may be so because they encode partly distinct components of the operating point, providing a quantified answer to the complementarity question posed in Section 2.

Which single channel is preferable proved to be model-family dependent: Elastic Net favours US while LightGBM favours AE, both significantly. As shown in Section 7.2, Elastic Net recovers a substantially larger share of the US channel’s tree-model performance than of AE’s. The AE feature– κ mapping is markedly less accessible to a linear model, consistent with AE resolving κ primarily through rotational speed, whose influence on κ is structurally non-linear (Eq. (2)). Channel rankings obtained under a single model family should therefore be treated with caution.

The comparison is further bounded by hardware. The AE sensor captures wideband content up to 1 MHz, dominated by verified stress-wave emission, whereas the US probe down-mixes a narrow ≈ 38 –40 kHz sensing band to an audio-frequency baseband, within which the 10–20 kHz band carries the strongest correlations. The high-frequency transient signature of individual contact events is largely absent from this down-mixed envelope. Whether a wideband passive ultrasound sensor

would supply more independent information and a larger fusion gain remains an open question.

8.2 On the use of κ as lubrication reference

The results indicate that the acoustic features are informative about lubrication conditions as quantified by κ . From a single externally mounted sensor, without tachometer or thermometer access, the models recover the bearing’s nominal lubrication adequacy—a capability of direct value for retrofit monitoring. This claim is, however, subject to the limitations of κ itself.

κ is not measured but computed for each segment from the measured operating point via the ISO 281 relations and the ASTM D341 Walther equation (Section 2.1). It therefore inherits the assumptions of those relations, notably the use of housing temperature as a proxy for lubricant temperature and the assumption of nominally intact lubricant. A further assumption is that lubricant supply is sufficient for fully flooded conditions. As noted there, high rotational speeds can induce starvation that reduces the actual film below κ ’s prediction even when the lubricant is intact. Whether the staircase protocol’s higher speed steps induced starvation, and whether the acoustic signals carry a starvation signature that the regression attributes to the operating point rather than the film state, has not been investigated. The sampled κ distribution is also concentrated in the boundary and mixed regimes; extrapolation into the full-film range ($\kappa > 1$) should be treated with caution.

More fundamentally, because κ is deterministic in the operating point, some degree of κ -regression success is guaranteed for any feature set that is informative about speed and temperature. The conditional correlation analysis and the clustered SHAP interpretation indicate that the models indeed draw primarily on operating-point-tracking features: in performance terms, the regression acts as an acoustic operating-point soft sensor combined with the ISO 281 mapping.

In addition, since κ encodes only lubrication changes caused by the operating point (Section 2.1), sensitivity to degradation, contamination, or starvation is neither demonstrated nor ruled out in this study or in prior κ -estimation work such as that of Jakobsen et al. Probing lubrication information beyond the operating point requires an independent reference, e.g. a film-thickness measurement, electrical contact resistance, or an experiment in which the lubricant state is varied independently of the operating point.

The complementarity conclusion is unaffected by these limitations since both channels are evaluated against the same target under identical protocols. The demonstrated gain from combining them reflects genuinely non-redundant information.

8.3 Evaluation protocol and comparability with prior work

The acquisition-grouped, twin-merged evaluation removes the near-duplicate leakage admitted by random splits over short measurement segments, and the group-level DM tests avoid the potential significance inflation of segment-level testing. The reported figures are consequently not directly comparable to prior work evaluated under random splits, including that of Jakobsen et al.; they are intended as more conservative, deployment-representative estimates.

A caveat applies in the opposite direction as well. Because the (rotational speed, temperature) space is densely and smoothly sampled, every held-out group is surrounded by near-neighbour training points: the high R^2 reflects *interpolation* of the smooth κ (operating-point) surface, consistent with the soft-sensor interpretation, rather than extrapolation to unseen regimes. Same-operating-point leakage is controlled, but generalisation to unsampled operating regimes is untested.

Finally, all results are specific to the SYJ25 bearing, Keratech 22 oil, and one 14-hour experimental session. Whether these conclusions hold for other bearing types, sizes, lubricants, or installations is unknown.

9 Conclusion

This paper has presented a controlled study in which wideband acoustic emission and heterodyned passive ultrasound were acquired simultaneously from a ball bearing across speed sweeps and temperature blocks, and regressed to the ISO 281 viscosity ratio κ under an acquisition-grouped, leakage-controlled evaluation protocol. The main findings are as follows:

1. **AE and passive US are complementary.** The combined feature set outperforms either channel alone for both model families, significantly in all four group-level contrasts. The complementarity has a conditional correlation basis: both channels track rotational speed, but US is more strongly associated with temperature than AE, so the two channels encode partly distinct components of the operating point. The clustered SHAP analysis confirms this at the feature level, with the combined model retaining temperature-leaning US features while discarding US amplitude features whose speed information the AE features already provide. Which single channel performs best is model-family dependent, so single-model channel comparisons should be interpreted with caution.
2. **The acoustic features are informative about lubrication conditions as quantified by κ .** Either channel alone supports κ regression well, recovering the bearing's nominal lubrication adequacy from externally mounted sensors without tachometer or thermometer access. Because κ is determined by the operating point, and the feature and SHAP analyses indicate that the models draw primarily on operating-point-tracking features, this accuracy could be understood as operating-point soft sensing. Sensitivity to lubrication changes not caused by the operating point, such as degradation, contamination, starvation, is neither demonstrated nor ruled out, and remains an open problem.
3. **The evaluation is leakage-controlled and conservative.** Grouping segments by acquisition hold and operating-point similarity prevents near-duplicate segments from crossing the training–evaluation boundary, and group-level significance testing avoids the inflation of segment-level tests. The reported figures are conservative estimates.

Taken together, the study provides the first quantitative demonstration of AE–US complementarity for lubrication condition monitoring, enabled by simultaneous, time-synchronised acquisition from the same bearing. Future work should test whether a wideband passive ultrasound channel widens the modest fusion gain found here, whether the complementarity persists across bearing types, lubricants, and installations, and whether the acoustic signals carry lubrication information beyond the operating point. This would require decoupling the lubricant state from the operating conditions, for example through an independent film reference or controlled degradation.

CRedit authorship contribution statement

Frederik Appel Vardinghus-Nielsen: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Resources, Software, Validation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interests

The author declares that there are no competing interests.

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Data availability

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Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used AI-assisted writing tools for language editing and structural drafting. After using these tools, the author reviewed and edited the content as necessary and takes full responsibility for the content of the publication.

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Public release of the complete dataset (HDF5 signal files) is not yet decided—resolve with CeramicSpeed and choose: (a) full Zenodo deposit, (b) features + code only, or (c) available from the corresponding author on reasonable request. Align contribution 4 and this statement with the decision.